Focus on the Core: Empowering Diffusion Large Language Models by Self-Contrast

Jinyuan Feng[1,2] **Xin Yu**[1] **Yiqun Chen**[3] **Xiaochi Wei**[4]
Yan Gao[4] **Yi Wu**[4] **Yao Hu**[4] **Zhiqiang Pu**[1,2]
[1]Institute of Automation, Chinese Academy of Sciences
[2]School of Artificial Intelligence, University of Chinese Academy of Sciences
[3]Renmin University of China
[4]Xiaohongshu Inc.

Abstract

The iterative denoising paradigm of Diffusion Large Language Models (DLMs) endows them with a distinct advantage in global context modeling. However, current decoding strategies fail to leverage this capability, typically exhibiting a local preference that overlooks the heterogeneous information density within the context, ultimately degrading generation quality. To address this limitation, we systematically investigate high-information-density (HD) tokens and present two key findings: (1) explicitly conditioning on HD tokens substantially improves output quality; and (2) HD tokens exhibit an early-decoding tendency, converging earlier than surrounding tokens. Motivated by these findings, we propose Focus on the Core **(FoCore)**, a training-free decoding strategy that utilizes HD tokens in a self-contrast manner, wherein HD tokens are temporarily remasked as negative samples, to guide generation. We further introduce FoCore_Accelerate **(FoCore_A)**, an efficient variant that, upon detecting HD token convergence, performs parallel decoding over stable candidates within a local context window, substantially accelerating generation. Extensive experiments on math, code and logical reasoning benchmarks demonstrate that FoCore consistently improves generation quality and efficiency across both LLaDA and Dream backbones. For instance, on HumanEval, FoCore improves pass@1 from 39.02 to 42.68 over standard Classifier-Free Guidance, while FoCore-A reduces the number of decoding steps by 2.07× and per-sample latency from 20.76s to 8.64s (-58.4%).

1 Introduction

By replacing sequential generation with iterative denoising, diffusion large language models (DLMs) present an alternative to autoregressive (AR) architectures, enabling global context modeling and flexible decoding [Nie et al., 2025, Ye et al., 2025]. However, this advantage is frequently bottlenecked by locally-biased decoding strategies: candidate tokens in closer proximity to already-denoised positions tend to receive disproportionately higher confidence scores [Fang et al., 2026, Cai and Li, 2026]. This locally-biased paradigm fundamentally neglects the heterogeneous information density inherent in natural language, as it confines the model's attention to a narrow positional neighborhood, failing to capture core reasoning tokens within the global context [Ma et al., 2026]. *Therefore, rather than passively sampling high-confidence tokens locally, proactively prioritizing core reasoning units is essential to fully unleash the contextual capabilities of DLMs.*

In this paper, we explore how core reasoning tokens influence generation quality through a novel perspective of "Heterogeneous Information Density" within DLMs. Our analysis reveals a significant heterogeneity in the information density of contextual tokens, where the majority of ordinary tokens

Preprint.

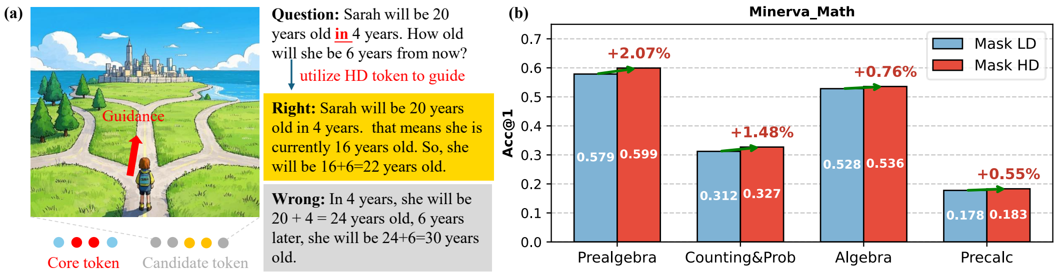


Figure 1: (a) A qualitative example illustrating that focusing on HD tokens effectively guides correct reasoning generation. (b) Quantitative results on Minerva_Math show that conditioning on HD tokens (**Mask HD**) consistently outperforms LD tokens (**Mask LD**).

function merely as redundant background padding, designated as *"low-information-density (LD)" tokens*, whereas a minority of core tokens act as logical anchors with exceptionally *"high-information-density", termed HD tokens*. During the denoising processes of DLMs, HD tokens serve as pivotal anchors that steer the reasoning toward correct trajectories. As illustrated by the qualitative analysis in Figure 1(a), HD tokens effectively constrain the denoising process when the model encounters divergent reasoning paths. For instance, in the depicted age calculation problem, the temporal preposition "in" *("Sarah will be 20 years old in 4 years")* functions as a quintessential HD token. This HD token dictates the core logical branch for deriving the current age, such that correctly capturing it steers the model toward the correct path *(identifying the current age as 16 to yield the final answer of 22)*, whereas neglecting it risks misguiding the reasoning into an erroneous addition *(e.g., 20+4=24)*.

To quantitatively validate the positive impact of HD tokens, we designed comparative experiments as depicted in Figure 1(b). Specifically, we evaluated the performance disparity between a decoding strategy prioritizing HD tokens (Mask HD) and an alter prioritizing LD tokens (Mask LD). Here, the prioritization of tokens is achieved via Classifier-Free Guidance [Ho and Salimans, 2022], which amplifies the model's sensitivity by contrasting conditional and unconditional denoising trajectories. The results indicate that the Mask HD consistently outperformed Mask LD in terms of accuracy (Acc@1) across all mathematical sub-domains. Notably, this strategy achieved a maximum accuracy increase of +2.07% (from 0.579 to 0.599) in the Prealgebra task and a +1.48% improvement in Counting & Probability, while maintaining stable positive gains even in the more challenging Algebra (+0.76%) and Precalculus (+0.55%) tasks. In essence, it is HD tokens, rather than LD counterparts, that constitute the informational backbone governing complex reasoning trajectories, underscoring that selectively prioritizing high-density information is effective for optimizing generation quality.

Motivated by the aforementioned validations, we propose **FoCore** (**Fo**cus on the **Core**), a training-free decoding strategy that leverages HD tokens in a self-contrast manner. Specifically, FoCore dynamically identifies and temporarily remasks converged HD tokens throughout the denoising trajectory to construct contrastive negative samples, and steers the generation by contrasting the output logits between the original and masked conditions, thereby compelling the model to strictly anchor its attention on core HD tokens during subsequent denoising steps. We further dive deep into the decoding dynamics of HD tokens throughout the denoising process, revealing an intriguing property: *HD tokens exhibit an early-decoding tendency, converging to their final states earlier than surrounding tokens*. Motivated by this finding, we introduce **FoCore_A**, an efficient variant that, upon detecting HD token convergence, performs parallel decoding over stable candidates within a local context window, substantially accelerating generation. Our contributions are threefold:

- **Empirical analysis on HD tokens:** We systematically investigate the behavioral properties of HD tokens and derive three critical insights that collectively demonstrate their dual benefits in generation quality and decoding efficiency, serving as the motivation and supporting evidence for the design of FoCore and FoCore_A.

- **A decoding strategy leveraging HD tokens:** We propose FoCore, a training-free decoding strategy that dynamically remasks converged HD tokens as negative samples to enhance generation quality, and FoCore_A, an efficient variant that exploits their early-decoding tendency for parallel decoding to further accelerate generation.

- **Comprehensive experimental evaluation:** Extensive experiments on various benchmarks demonstrate that FoCore consistently enhances generation quality across both LLaDA and Dream backbones, while FoCore_A reduces per-sample latency from 20.76s to 8.64s (-58.4%) with a $2.07\times$ decoding step reduction over standard Classifier-Free Guidance.

2 An Exploratory Analysis on DLMs

2.1 Preliminary on DLMs

DLMs formulate text generation as an iterative denoising process, centered on two probabilistic processes applied to a clean sequence $x_0 \sim p_{\text{data}}(x_0)$ of length L: a forward noising process that progressively corrupts x_0 into noise, and a reverse denoising process that learns to reconstruct x_0 from the corrupted input. Due to space constraints, additional related work is provided in Appendix A.

Forward Noising Process. In the discrete space, the forward process is a Markov chain progressively introducing an absorbing state, `[MASK]` (denoted as m) [Austin et al., 2021a]. By unrolling the step-wise transitions $q(x_t|x_{t-1})$, the marginal probability at any step $t \in [0, T]$ is analytically condensed into a categorical distribution governed by a monotonically decreasing schedule α_t:

$$q(x_{1:T}|x_0) = \prod_{t=1}^{T} q(x_t|x_{t-1}) \quad \Longrightarrow \quad q(x_t|x_0) = \text{Cat}(x_t; \alpha_t x_0 + (1-\alpha_t)m). \tag{1}$$

This indicates that at step t, each token retains its original identity x_0 with probability α_t, and is corrupted into m with probability $1 - \alpha_t$. As $t \to T$, x_T degenerates into a fully masked state.

Reverse Generative Process. The reverse process employs a neural network parameterized by θ to invert this corruption. Starting from a fully masked input $x_T = (m, \ldots, m)$, the iterative generation is rigorously formulated as:

$$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1}|x_t) = p(x_T) \prod_{t=1}^{T} q(x_{t-1}|x_t, \hat{x}_0) p_\theta(\hat{x}_0|x_t). \tag{2}$$

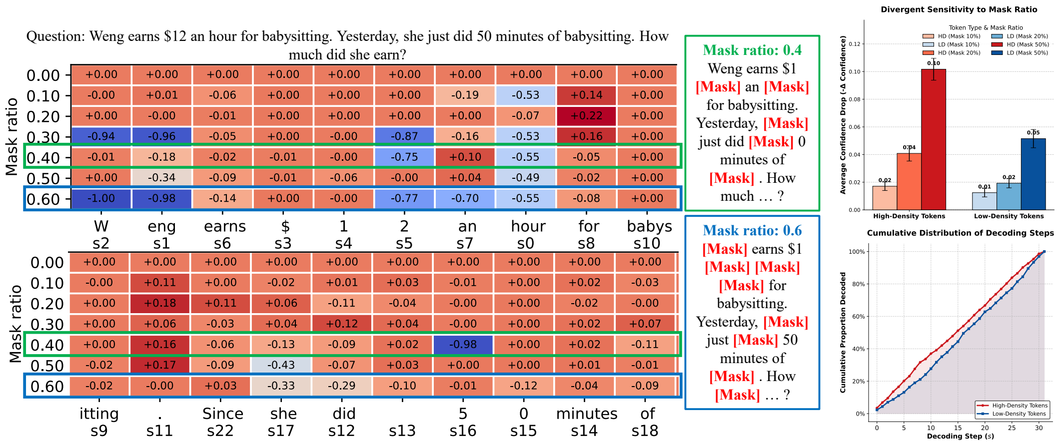
This formulation decomposes decoding into two steps: **(1) Prediction** $p_\theta(\hat{x}_0|x_t)$, where the model predicts a clean candidate $\hat{x}_0$ based on the noisy x_t; and **(2) Re-masking** $q(x_{t-1}|x_t, \hat{x}_0)$, where the forward transition is re-applied to inject noise into $\hat{x}_0$. In practice, guided by heuristic strategies (e.g., predictive confidence) [Chang et al., 2022], the re-masking step selectively unmasks and fixes high-certainty tokens while replacing the rest with m. By iteratively alternating these two steps (often accelerated via step-skipping), DLMs eschew traditional autoregression, progressively reducing the mask ratio in a globally parallel manner to yield the coherent text x_0.

2.2 Decoding Dynamics of HD Tokens

To systematically characterize the decoding dynamics of HD tokens, we conduct qualitative and quantitative analyses (Figure 2) across three critical dimensions: information density, global context dependency, and decoding timing.These dimensions collectively reveal how HD tokens behave distinctly from other tokens throughout the iterative generation process, yielding three key findings:

Key Findings

- **Heterogeneous information density:** When the global context is partially masked, confidence drops (Δ Confidence) exhibit marked heterogeneity across token types. HD tokens (e.g., entities, numbers) suffer extreme collapses, establishing them as irreplaceable semantic hubs, whereas structural LD tokens remain robust.
- **Global context dependency:** The confidence of these core tokens degrades rapidly and non-linearly even under minimal global context masking, proving their strict reliance on a holistic sequence view to be accurately determined.
- **Propensity for early decoding:** Confidence-based decoding strategies inherently identify and denoise HD tokens at earlier generation steps. Rather than a structural flaw, this early-decoded HD tokens provides critical *semantic anchors* that can be leveraged to significantly accelerate the subsequent decoding of surrounding LD tokens.



(a) (b)

Figure 2: Overview of the decoding characteristics of HD tokens: high contextual sensitivity and premature commitment. **(a) Case Study:** Heatmaps show the confidence drop (Δ Confidence) under varying global mask ratios. Highlighted HD tokens (e.g., numbers, entities) suffer severe confidence collapse without their corresponding HD tokens in the global context. **(b) Quantitative Analysis:** The top chart proves that HD tokens are significantly more sensitive to context masking than LD tokens. Counterintuitively, the bottom curve reveals that current strategies prematurely decode these highly sensitive HD tokens at earlier steps. Detailed analysis is provided in Section 2.2.

To substantiate these findings with rigorous empirical grounding, we conduct a systematic and in-depth analysis across each identified dimension:

Analysis of Heterogeneous Information Density (Refining Finding 1): To construct the heatmap in Figure 2a, we systematically obscure the input prompt at varying mask ratios (y-axis) and record the corresponding confidence drops (Δ Confidence) for each generated token. As visualized, this perturbation reveals a striking dichotomy in information density: HD tokens (e.g., "W", "2", "hour") suffer extreme, deep-blue confidence collapses (approaching -1.00) when deprived of global context, whereas structural LD tokens (e.g., "for", "she") remain robustly unaffected (orange cells). *Crucially, examining specific perturbation scenarios (mask ratios of* 0.4 *and* 0.6*, green and blue boxes) exposes a profound intra-sequence dependency.* The localized variations in the heatmap strictly align with which specific prompt tokens are masked. For instance, under the 0.4 mask ratio, obscuring the HD token "5" (from "50 minutes") in the prompt causes the model's predictive confidence for the corresponding target HD token "5" (s_{16}) to catastrophically plummet to -0.98. This confirms that high-density semantic hubs are intricately coupled; perturbing even a single HD token in the context catastrophically impairs the prediction of related HD tokens.

Analysis of Global context dependency: (Refining Finding 2): Figure 2b (**Top**) quantifies sequence-level heterogeneous information density (500 samples) by contrasting the confidence drop ($-\Delta$ Confidence) of HD tokens (e.g., entities, nums) versus LD tokens (e.g., prepositions) across varying global mask ratios. The results demonstrate a distinctly divergent sensitivity: HD tokens suffer rapid, non-linear degradation ($0.02 \rightarrow 0.04 \rightarrow 0.10$), culminating at 50% masking with a confidence drop (0.10) precisely double that of robust LD tokens (0.05). This statistically substantiates the *strict global context dependency* inherent in DLMs. It proves that accurately predicting core semantic hubs dictates an entirely uncorrupted holistic sequence view; thus, even minimal contextual perturbations disproportionately penalize HD tokens, inevitably triggering a cascade of subsequent semantic errors.

Analysis of Propensity for early decoding (Refining Finding 3): Figure 2b (**Bottom**) tracks the cumulative decoding distribution (s), exposing a systematic temporal divergence: driven by local high-confidence heuristics, the model intrinsically prioritizes core semantic hubs, causing HD tokens solidification to consistently outpace LD tokens (e.g., at $s = 10$, $\sim$38% HD vs. $\sim$28% LD decoded). Crucially, this innate "core-first" stabilization unlocks a paradigm for inference acceleration. Once the critical reasoning backbone (HD) is established early, applying computationally expensive iterative decoding to the structurally predictable LD residuals becomes redundant. Consequently, late-stage generation can be aggressively consolidated into a single parallel pass, drastically slashing latency without compromising semantic integrity.

3 Methodology

In this section, we present our methodology by answering two questions: (1) how can HD tokens be accurately identified, and (2) how can they be leveraged to improve performance and efficiency?

3.1 Identification of HD token

In our quantitative analysis conducted on Figure 1(b), we identify HD tokens by leveraging the SpaCy natural language processing library [Honnibal et al., 2020] in conjunction with part-of-speech tagging and named entity recognition. This identification mechanism is operationalized through a two-step procedure. Initially, during the text decoding and parsing phase, the token IDs generated by the large language model are decoded into plain text, stripped of leading and trailing whitespace, and processed through the SpaCy *en_core_web_sm* model for syntactic analysis. Subsequently, a dual-rule matching criterion is applied, which dictates that any text segment containing a noun, a proper noun, a number, or constituting a recognized named entity is classified as a HD token.

Figure 3: HD tokens in MATH

Figure 4: HD tokens in MBPP

	HumanEval	MBPP	BBH
Mask HD	38.41	35.00	56.12
Mask LD	36.50	36.60	57.02
JS_detect	**42.68**	**40.40**	**58.37**

Table 1: Performance Comparison

While this linguistic-rule-based heuristic approach demonstrates significant efficacy in extracting HD tokens within the MATH dataset, our subsequent evaluation across the code generation and reasoning benchmarks (HumanEval, MBPP, and BBH), reveal its inherent limitations. Specifically, as illustrated in Figure 3 and 4, while HD tokens in the MATH dataset align naturally with linguistic rules (e.g., nouns like `days` and numbers like `7`, `5`), those in code generation benchmarks such as MBPP exhibit substantially different characteristics, encompassing programming keywords (e.g., `def`, `return`, `sort`) and code-specific identifiers (e.g., `string`, `lowercase`, `Counter`). Such distinction renders the linguistic-rule-based heuristic fundamentally inadequate, exposing its over-reliance on human priors and excessive domain coupling. To this end, achieving domain-agnostic HD token identification necessitates a paradigm shift from externally defined rule sets toward the model's intrinsic perceptual capabilities, motivating us to propose a fully automated approach.

Automated Identification via Distribution Stability The underlying assumption is that if a decoded token is an HD token, the model's predicted probability distribution for that position should exhibit significant divergence across consecutive iterations. As empirically evidenced in Figure 2b (**Top**) and consistent with [Zou et al., 2026], HD tokens tend to induce notable distributional shifts across decoding steps, whereas LD tokens maintain a relatively stable predictive distribution.

Let $P_t(x_i)$ denote the predicted probability distribution over the vocabulary for the i-th position at iteration t. To quantify the discrepancy between the predictions at step t and $t-1$, we employ an approximated Jensen-Shannon (JS) divergence. Considering the computational overhead of calculating the divergence over the entire vocabulary and the noise introduced by long-tail, low-frequency tokens, we propose a Top-K truncated JS divergence, which restricts the computation to the K most probable tokens at the current step. Formally, let $\mathcal{V}_K = \text{Top-}K\big(P_t(x_i)\big)$ denote the set of the K highest-probability tokens under $P_t(x_i)$. The step-wise instability $D_t^{(i)}$ for the i-th position at step t is then defined as:

$$D_t^{(i)} = \frac{1}{2} \sum_{v \in \mathcal{V}_K} P_t(v) \log\left(\frac{P_t(v)}{M(v)}\right) + \frac{1}{2} \sum_{v \in \mathcal{V}_K} P_{t-1}(v) \log\left(\frac{P_{t-1}(v)}{M(v)}\right) \tag{3}$$

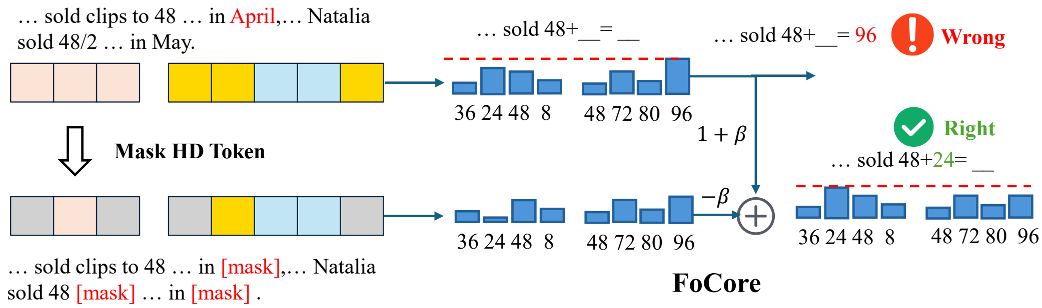


Figure 5: An illustration of the FoCore self-contrast mechanism. FoCore identifies and temporarily remasks HD tokens throughout the denoising trajectory to construct contrastive negative samples.

where $M(v) = \frac{1}{2}\big(P_t(v) + P_{t-1}(v)\big)$. A larger $D_t^{(i)}$ indicates a greater distributional shift at that position, suggesting that the corresponding token is more likely to be a HD token.

Temporal Smoothing via Historical Momentum During the iterative decoding process, the continuous updating of other masked tokens in the sequence inevitably introduces dynamic contextual shifts, which can cause step-wise fluctuations in distributions. Therefore, relying solely on the single-step divergence $D_t^{(i)}$ is prone to misjudgments. To obtain a robust evaluation metric, we introduce an Exponential Moving Average (EMA) to track the historical evolution of the distribution divergence. For each position i, we maintain a cumulative instability score $S_t^{(i)}$, updated as follows:

$$S_t^{(i)} = \alpha \cdot S_{t-1}^{(i)} + (1 - \alpha) \cdot D_t^{(i)} \tag{4}$$

where $\alpha \in [0, 1)$ is the EMA decay coefficient. Through this mechanism, $S_t^{(i)}$ encodes the memory of the token's stability over past iterations. Ultimately, a token with a high cumulative instability $S_t^{(i)}$ indicates that the model's predicted distribution for this position undergoes significant shifts throughout the iterative decoding process, suggesting a strong likelihood of being an HD token. As demonstrated in Table 1, our automated identification approach (**JS_detect**) achieves substantial improvements over the linguistic-rule-based heuristic, yielding gains of 4.27, 5.40, and 2.25 points on HumanEval, MBPP, and BBH, respectively.

3.2 Exploitation of HD Tokens

As illustrated in Figure 5, this subsection elaborates on how FoCore exploits HD tokens via Self-Contrast: FoCore dynamically identifies and temporarily remasks HD tokens throughout the denoising trajectory to construct contrastive negative samples, and subsequently steers the generation by contrasting the output logits between the original and masked conditions, thereby compelling the model to strictly anchor its attention on core HD tokens.

Construction of the negative samples At iteration t, let X_t denote the current sequence state, and $\mathcal{U}_t$ denote the set of indices for currently unmasked tokens. We aim to synthesize a "unconditional" input sequence, denoted as $\tilde{X}_t$, by selectively masking the HD tokens of the sequence.

First, using the cumulative instability scores $S_t^{(i)}$ computed previously, we identify the top-K unstable tokens within the decoded sequence. Specifically, we select a subset of indices $\mathcal{K}_t \subset \mathcal{U}_t$ such that:

$$\mathcal{K}_t = \underset{\mathcal{K} \subset \mathcal{U}_t, |\mathcal{K}| = K}{\arg\max} \sum_{j \in \mathcal{K}} \left(S_t^{(j)} + \epsilon \right) \tag{5}$$

where $\epsilon \sim \mathcal{U}(0, 10^{-8})$ is an infinitesimal uniform noise added to break ties. Subsequently, we construct the unconditional input $\tilde{X}_t$ by modifying the current sequence X_t. For the non-`[MASK]` tokens whose indices belong to $\mathcal{K}_t$, we explicitly replace them with the special `[MASK]` token, while the rest of the sequence (including the stable tokens and the masked tokens) remains unchanged:

$$(\tilde{X}_t)_j = \begin{cases} \texttt{[MASK]}, & \text{if } j \in \mathcal{K}_t \\ (X_t)_j, & \text{otherwise.} \end{cases} \tag{6}$$

Note that when all instability scores are equal (e.g., $S_t^{(i)} = 0$ at early iterations), the selection of $\mathcal{K}_t$ is governed solely by ϵ, reducing the mechanism to random masking.

Self-Contrast Mechanism Given the current sequence state X_t and the negative sample $\tilde{X}_t$, we perform two forward passes through f_θ to obtain:

$$L_{\text{cond}} = f_\theta(X_t), \quad L_{\text{uncond}} = f_\theta(\tilde{X}_t) \tag{7}$$

The self-contrastive logits $\hat{L}_t$ are then formulated as:

$$\hat{L}_t = L_{\text{uncond}} + (\omega + 1) \cdot \big(L_{\text{cond}} - L_{\text{uncond}}\big) \tag{8}$$

where $\omega > 0$ is the guidance scale, which is mathematically equivalent to $\hat{L}_t = L_{\text{cond}} + \omega \cdot \big(L_{\text{cond}} - L_{\text{uncond}}\big)$. The resulting $\hat{L}_t$ subsequently replaces the original logits in the decoding step, where tokens are sampled based on the self-contrastively guided distribution.

FoCore_A variants We further introduce **FoCore_A**, an efficient variant that accelerates inference via parallel decoding. Specifically, FoCore_A continuously monitors the mean EMA-based instability score over undecoded positions, defined as $\bar{s}_i = \frac{1}{|\mathcal{U}_t|} \sum_{j \in \mathcal{U}_t} \text{EMA}_t^{(j)}$. Once $\bar{s}_i < \tau$, indicating sufficient convergence of HD tokens within the current block, FoCore_A immediately decodes m additional high-confidence tokens in parallel at the current step, significantly reducing the total number of forward passes while preserving generation quality.

Pseudo-Code The pseudo-code for FoCore and FoCore_A is provided in Appendix C.

Theoretical Analysis Appendix D offers a theoretical analysis of how FoCore facilitates decoding.

4 Experiments

4.1 Experimental Setup

Datasets and Backbones To comprehensively evaluate the proposed approach, we conduct experiments across six widely adopted benchmarks covering mathematical reasoning, code generation, and logical reasoning tasks. Specifically, for mathematical reasoning, we utilize GSM8K [Cobbe et al., 2021] and MATH500 [Hendrycks et al., 2021]. For code generation, we employ HumanEval [Chen et al., 2021] and MBPP [Austin et al., 2021b]. For logical reasoning, we evaluate on SVAMP [Patel et al., 2021] and Countdown [Gandhi et al., 2024]. All experiments are conducted using two robust instruction-tuned large language models as our backbones: LLaDA 8B Instruct [Nie et al., 2025] and Dream 7B Instruct [Ye et al., 2025]. Further implementation details are provided in Appendix B.

Baselines We benchmark our proposed method against a diverse set of strong baselines to demonstrate its effectiveness. These baselines can be broadly categorized into two groups. The first group consists of standard and widely used decoding strategies, including Random sampling, Greedy decoding, Confidence-based decoding, and Standard Classifier-Free Guidance (Std CFG). The second group comprises recent state-of-the-art (SOTA) advanced decoding methods, specifically A-CFG [Li et al., 2025a], KLASS [Kim et al.], and Prophet [Li et al., 2025b].

4.2 Main Results

Effectiveness. As illustrated in Table 4, FoCore achieves systematic performance gains over all baselines on both Math and Code tasks under the LLaDA-8B-Instruct backbone. On mathematical benchmarks, FoCore attains 75.05 on GSM8K and 43.80 on MATH500, surpassing the strongest baselines KLASS (74.14) and Confidence (42.60), respectively. The improvements are particularly pronounced on code generation: FoCore yields gains of +3.1 to +3.7 percentage points over the strongest baselines on HumanEval, and consistently ranks first across all generation lengths on MBPP (38.40/40.40/40.00), uniformly outperforming all competitors including Prophet. As a lightweight variant, FoCore_A incurs only marginal performance degradation while remaining competitive across most configurations. Results on SVAMP and Countdown are reported in Appendix E.

Scalability. FoCore exhibits a stable and monotonically increasing performance trend as the generation length scales from 128 to 512, in contrast to the unstable scaling behavior observed in most baselines. On HumanEval, FoCore consistently improves across lengths (36.58→42.68→44.51),

Dataset / Methods / Len	GSM8K (Math) 128	256	512	MATH500 (Math) 128	256	512	HumanEval (Code) 128	256	512	MBPP (Code) 128	256	512
LLaDA-8B-Instruct												
Random	63.76	67.93	69.14	26.20	28.60	27.00	18.29	23.17	22.56	26.60	28.20	27.00
Greedy	74.07	72.70	74.07	32.20	42.40	39.20	27.43	37.80	36.58	36.80	36.20	36.80
Confidence	74.07	71.03	74.07	32.80	42.60	38.60	27.43	37.80	36.58	37.40	36.60	37.20
Std CFG	71.64	71.72	73.16	34.20	33.40	35.60	33.53	39.02	38.41	36.20	39.40	38.40
A-CFG [NeurIPS 25]	71.11	70.43	72.32	33.20	38.20	37.60	26.82	31.09	41.46	34.40	36.00	37.40
KLASS [NeurIPS 25]	74.14	71.19	74.37	32.80	42.20	38.60	27.43	37.80	37.19	37.00	37.00	37.20
Prophet [ICLR 26]	–	69.44	–	–	31.20	–	–	36.58	–	–	39.80	–
FoCore (ours)	**74.67**	**75.05**	74.45	32.80	**43.80**	39.00	**36.58**	**42.68**	**44.51**	**38.40**	**40.40**	**40.00**
FoCore_A (ours)	73.46	73.99	**75.44**	**34.80**	42.20	**39.60**	28.04	40.24	40.85	36.60	36.20	39.80
Dream-Instruct-7B												
Random	35.10	37.60	38.13	14.20	14.00	16.80	18.90	22.50	21.30	29.40	30.80	32.40
Greedy	75.11	80.66	79.22	40.00	38.00	42.20	52.43	53.65	50.00	55.20	56.80	57.00
Confidence	76.19	79.52	79.22	41.00	39.80	43.40	56.09	53.65	49.39	55.20	56.60	57.40
Std CFG	75.13	80.06	79.45	40.60	40.00	42.00	**56.70**	52.44	52.43	54.60	58.00	57.20
A-CFG [NeurIPS 25]	73.84	80.21	78.01	40.40	42.80	40.60	53.04	51.82	48.17	56.20	57.80	57.40
KLASS [NeurIPS 25]	76.04	79.37	80.13	39.40	39.00	43.80	53.66	51.21	52.43	54.40	56.80	57.60
Prophet [ICLR 26]	–	72.25	–	–	39.60	–	–	51.80	–	–	57.60	–
FoCore (ours)	75.43	**80.97**	**81.11**	**43.40**	**43.80**	**44.40**	**56.70**	**54.87**	**53.04**	**56.60**	**58.20**	**59.80**
FoCore_A (ours)	**76.64**	80.59	79.15	40.40	41.80	**44.40**	54.87	53.65	53.04	**56.60**	58.00	56.20

Table 2: Quantitative comparison of FoCore and FoCore_A against baselines on math (GSM8K, MATH500) and code (HumanEval, MBPP) benchmarks under the LLaDA-8B-Instruct and Dream-Instruct-7B backbones, with generation lengths $\{128, 256, 512\}$. For each column, the best result within each backbone block is highlighted in red and the second-best in green; the column-wise maximum is additionally marked in **bold**.

making it the only method that outperforms all baselines at every length; a similar monotonic trend holds on GSM8K (74.67→75.05), while Greedy and Confidence suffer a noticeable drop at length 256 (72.70), and A-CFG only reaches its peak at length 512 (41.46). These results collectively demonstrate that FoCore makes more effective use of longer generation budgets. FoCore_A likewise exhibits robust scalability, achieving the column-best score of 75.44 on GSM8K at length 512, with only marginal underperformance relative to FoCore at certain intermediate lengths.

Generalizability. FoCore maintains its dominant performance when transferred to the Dream backbone, demonstrating strong generalizability across discrete diffusion language model architectures. On MATH500, FoCore attains column-best results across all three generation lengths (43.40/43.80/44.40), comprehensively surpassing all baselines. On code generation, FoCore achieves 59.80 on MBPP at length 512, outperforming the strongest baseline KLASS (57.60) by +2.2 percentage points, and consistently outperforms or matches all baselines on HumanEval. FoCore_A similarly suffers only negligible performance degradation, collectively demonstrating the robust generalizability of the proposed framework across the family of discrete diffusion language models.

Computational Complexity. Beyond performance, FoCore_A achieves favorable efficiency by leveraging EMA-based instability monitoring to trigger parallel decoding, which significantly reduces the total number of forward passes. As shown in Table 3(a), FoCore_A attains 1.45× to 2.29× decoding step speedup across all benchmarks, while simultane-

Dataset / Method	GSM8K Perf.↑	Speedup↑	MATH500 Perf.↑	Speedup↑	HumanEval Perf.↑	Speedup↑	MBPP Perf.↑	Speedup↑
(a) Decoding Step Acceleration								
Prophet	69.44	1.33×	31.20	1.25×	36.58	1.26×	**39.80**	1.24×
KLASS	71.19	**1.90**×	42.20	**2.13**×	37.80	**2.42**×	37.00	1.45×
FoCore_A	**73.99**	1.45×	**42.20**	1.88×	**40.24**	2.07×	36.20	**2.29**×

Dataset / Method	GSM8K Perf.↑	Time(s)↓	MATH500 Perf.↑	Time(s)↓	HumanEval Perf.↑	Time(s)↓	MBPP Perf.↑	Time(s)↓
(b) End-to-End Generation Efficiency								
A-CFG	70.43	19.85	38.20	21.38	31.09	22.56	36.00	19.27
CFG	71.72	16.61	33.40	18.28	39.02	20.76	**39.40**	15.49
FoCore_A	**73.99**	**12.65**	**42.20**	**11.56**	**40.24**	**8.64**	36.20	**7.45**

Table 3: Efficiency comparison across four benchmarks. **(a)** Decoding step speedup (×) over standard decoding, paired with task performance (%). **(b)** End-to-end generation time (s), paired with task performance (%). ↑ higher is better; ↓ lower is better. Bold denotes best per column.

ously outperforming competing efficient methods Prophet and KLASS in task performance on GSM8K (73.99 vs. 71.19) and HumanEval (40.24 vs. 37.80). In terms of end-to-end generation time (Table 3(b)), FoCore_A achieves the lowest wall-clock time across nearly all benchmarks, reducing generation time by up to $2.4\times$ compared to A-CFG and $1.3\times$ compared to Std CFG, while maintaining superior task performance. These results demonstrate that FoCore_A strikes a favorable trade-off between efficiency and effectiveness, making it a practical choice for inference under computational constraints. A detailed ablation study on the parallel decoding hyperparameter m is provided in Appendix E.2.

4.3 Ablation Study and Analysis

Effect of ω As illustrated in Figure 6(a), the model demonstrates strong robustness to guidance scale across both datasets, maintaining stable performance over a broad range of values. The guidance scale fundamentally governs the trade-off between exploration and exploitation during the search process: an excessively small scale provides insufficient guidance signals, hindering the model from effectively concentrating on high-quality candidates, while an overly large scale over-constrains the search space diversity and risks converging to local optima. It is also noteworthy that performance on the Countdown task exhibits greater sensitivity to this hyperparameter than on SVAMP, which aligns with its nature as a combinatorial search task that inherently demands a wider exploration space. Overall, a guidance scale of approximately 0.3 is recommended, as it strikes an optimal balance between guidance strength and search diversity.

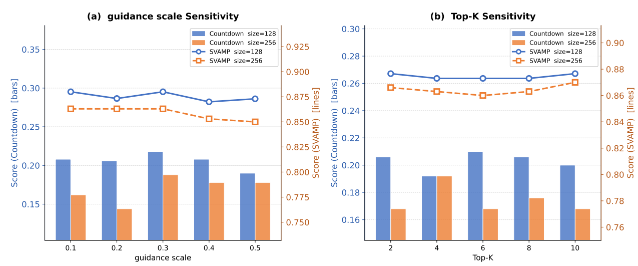


Figure 6: Hyperparameter sensitivity analysis of FoCore with respect to two key hyperparameters ω and Top-K on Countdown and SVAMP benchmarks at generation lengths 128 and 256. **(a)** Guidance scale $\omega \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$. **(b)** Top-$K \in \{2, 4, 6, 8, 10\}$.

Effect of Top-K As illustrated in Figure 6(b), the model exhibits minimal performance fluctuation across both datasets with respect to Top-K, demonstrating considerable robustness to this hyperparameter. This observation implies that HD Tokens are inherently sparse during the decoding process, such that even at relatively small values of K, the critical candidates are sufficiently covered and the model is capable of sustaining stable performance. However, as K grows substantially, the incorporation of excessive noisy candidates results in marginal performance degradation, suggesting that indiscriminately enlarging the candidate set does not consistently yield improvements. Furthermore, the optimal choice of K is found to interact with task-specific characteristics: for tasks characterized by a more open solution space, moderately increasing K promotes broader candidate coverage and proves beneficial, whereas for tasks with a more concentrated solution space, a smaller K already suffices to capture the most essential candidates, and an overly large candidate pool instead introduces unwanted interference.

Case Study of FoCore In Appendix F, we provide a qualitative analysis of FoCore, wherein the HD tokens during the decoding process are highlighted to demonstrate how FoCore leverages these anchors to guide the model toward the correct path.

5 Conclusion

This work uncovers a critical yet long-overlooked property of DLMs: the heterogeneous information density distribution inherent in the generated context. Through systematic investigation of high-information-density (HD) tokens, we demonstrate their central role in both semantic guidance and decoding acceleration. Specifically, FoCore steers generation by exploiting HD tokens in a training-free, self-contrastive manner, while FoCore_A further leverages their convergence behavior to substantially reduce decoding steps and per-sample latency. Extensive experiments across multiple benchmarks validate the effectiveness and efficiency of the proposed framework. Looking forward, we believe that incorporating HD token awareness into RL-based post-training of DLMs represents a promising direction for internalizing the focus-on-core principle at the parameter level.

References

Jacob Austin, Daniel D Johnson, Jonathan Ho, Daniel Tarlow, and Rianne Van Den Berg. Structured denoising diffusion models in discrete state-spaces. *Advances in neural information processing systems*, 34:17981–17993, 2021a.

Jacob Austin, Augustus Odena, Maxwell Nye, Maarten Bosma, Henryk Michalewski, David Dohan, Ellen Jiang, Carrie Cai, Michael Terry, Quoc Le, et al. Program synthesis with large language models. *arXiv preprint arXiv:2108.07732*, 2021b.

Eden Avrahami and Eliya Nachmani. Ilrr: Inference-time steering method for masked diffusion language models. *arXiv preprint arXiv:2601.21647*, 2026.

Tiwei Bie, Maosong Cao, Kun Chen, Lun Du, Mingliang Gong, Zhuochen Gong, Yanmei Gu, Jiaqi Hu, Zenan Huang, Zhenzhong Lan, et al. Llada2. 0: Scaling up diffusion language models to 100b. *arXiv preprint arXiv:2512.15745*, 2025.

Changxiao Cai and Gen Li. Confidence-based decoding is provably efficient for diffusion language models. *arXiv preprint arXiv:2603.22248*, 2026.

Mingyu Cao, Alvaro Correia, Christos Louizos, Shiwei Liu, and Lu Yin. Search or accelerate: Confidence-switched position beam search for diffusion language models. In *ICLR 2026 2nd Workshop on Deep Generative Model in Machine Learning: Theory, Principle and Efficacy*.

Michael Cardei, Jacob K Christopher, Bhavya Kailkhura, Thomas Hartvigsen, and Ferdinando Fioretto. Constrained discrete diffusion. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*.

Huiwen Chang, Han Zhang, Lu Jiang, Ce Liu, and William T Freeman. Maskgit: Masked generative image transformer. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 11315–11325, 2022.

Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde De Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, et al. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*, 2021.

Tianlang Chen, Minkai Xu, Jure Leskovec, and Stefano Ermon. Rfg: Test-time scaling for diffusion large language model reasoning with reward-free guidance. *arXiv preprint arXiv:2509.25604*, 2025.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

Meihua Dang, Jiaqi Han, Minkai Xu, Kai Xu, Akash Srivastava, and Stefano Ermon. Inference-time scaling of diffusion language models with particle gibbs sampling. *arXiv preprint arXiv:2507.08390*, 2025.

Sander Dieleman, Laurent Sartran, Arman Roshannai, Nikolay Savinov, Yaroslav Ganin, Pierre H Richemond, Arnaud Doucet, Robin Strudel, Chris Dyer, Conor Durkan, et al. Continuous diffusion for categorical data. *arXiv preprint arXiv:2211.15089*, 2022.

Liancheng Fang, Aiwei Liu, Henry Peng Zou, Yankai Chen, Enze Ma, Leyi Pan, Chunyu Miao, Wei-Chieh Huang, Xue Liu, and Philip S Yu. Locally confident, globally stuck: The quality-exploration dilemma in diffusion language models. *arXiv preprint arXiv:2604.00375*, 2026.

Kanishk Gandhi, Denise Lee, Gabriel Grand, Muxin Liu, Winson Cheng, Archit Sharma, and Noah D Goodman. Stream of search (sos): Learning to search in language. *arXiv preprint arXiv:2404.03683*, 2024.

Shansan Gong, Shivam Agarwal, Yizhe Zhang, Jiacheng Ye, Lin Zheng, Mukai Li, Chenxin An, Peilin Zhao, Wei Bi, Jiawei Han, et al. Scaling diffusion language models via adaptation from autoregressive models. *arXiv preprint arXiv:2410.17891*, 2024.

Xiaochuang Han, Sachin Kumar, and Yulia Tsvetkov. Ssd-lm: Semi-autoregressive simplex-based diffusion language model for text generation and modular control. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11575–11596, 2023.

Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the math dataset. *arXiv preprint arXiv:2103.03874*, 2021.

Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. *arXiv preprint arXiv:2207.12598*, 2022.

Matthew Honnibal, Ines Montani, Sofie Van Landeghem, Adriane Boyd, et al. spacy: Industrial-strength natural language processing in python. 2020.

Zhanqiu Hu, Jian Meng, Yash Akhauri, Mohamed S Abdelfattah, Jae-sun Seo, Zhiru Zhang, and Udit Gupta. Accelerating diffusion language model inference via efficient kv caching and guided diffusion. *arXiv e-prints*, pages arXiv–2505, 2025.

Zemin Huang, Yuhang Wang, Zhiyang Chen, and Guo-Jun Qi. Don't settle too early: Self-reflective remasking for diffusion language models. *arXiv preprint arXiv:2509.23653*, 2025.

Seo Hyun Kim, Sunwoo Hong, Hojung Jung, Youngrok Park, and Se-Young Yun. Klass: Kl-guided fast inference in masked diffusion models. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*.

Cheuk Kit Lee, Paul Jeha, Jes Frellsen, Pietro Lio, Michael Samuel Albergo, and Francisco Vargas. Debiasing guidance for discrete diffusion with sequential monte carlo. *arXiv preprint arXiv:2502.06079*, 2025.

Pengxiang Li, Shilin Yan, Joey Tsai, Renrui Zhang, Ruichuan An, Ziyu Guo, and Xiaowei Gao. Adaptive classifier-free guidance via dynamic low-confidence masking. *arXiv preprint arXiv:2505.20199*, 2025a.

Pengxiang Li, Yefan Zhou, Dilxat Muhtar, Lu Yin, Shilin Yan, Li Shen, Soroush Vosoughi, and Shiwei Liu. Diffusion language models know the answer before decoding. *arXiv preprint arXiv:2508.19982*, 2025b.

Tianyi Li, Mingda Chen, Bowei Guo, and Zhiqiang Shen. A survey on diffusion language models. *arXiv preprint arXiv:2508.10875*, 2025c.

Xiang Li, John Thickstun, Ishaan Gulrajani, Percy S Liang, and Tatsunori B Hashimoto. Diffusion-lm improves controllable text generation. *Advances in neural information processing systems*, 35: 4328–4343, 2022.

Justin Lovelace, Varsha Kishore, Yiwei Chen, and Kilian Weinberger. Diffusion guided language modeling. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 14936–14952, 2024.

Lizhuo Luo, Shenggui Li, Yonggang Wen, and Tianwei Zhang. Dsb: Dynamic sliding block scheduling for diffusion llms. *arXiv preprint arXiv:2602.05992*, 2026.

Linrui Ma, Yufei Cui, Kai Han, and Yunhe Wang. Mask is what dllm needs: A masked data training paradigm for diffusion llms. *arXiv preprint arXiv:2603.15803*, 2026.

Yichuan Mo, Quan Chen, Mingjie Li, Zeming Wei, and Yisen Wang. Decoding large language diffusion models with foreseeing movement. *arXiv preprint arXiv:2512.04135*, 2025.

Shen Nie, Fengqi Zhu, Zebin You, Xiaolu Zhang, Jingyang Ou, Jun Hu, Jun Zhou, Yankai Lin, Ji-Rong Wen, and Chongxuan Li. Large language diffusion models. *arXiv preprint arXiv:2502.09992*, 2025.

Arkil Patel, Satwik Bhattamishra, and Navin Goyal. Are nlp models really able to solve simple math word problems? In *Proceedings of the 2021 conference of the North American chapter of the association for computational linguistics: human language technologies*, pages 2080–2094, 2021.

Chitwan Saharia, William Chan, Saurabh Saxena, Lala Li, Jay Whang, Emily L Denton, Kamyar Ghasemipour, Raphael Gontijo Lopes, Burcu Karagol Ayan, Tim Salimans, et al. Photorealistic text-to-image diffusion models with deep language understanding. *Advances in neural information processing systems*, 35:36479–36494, 2022.

Wen Wang, Bozhen Fang, Chenchen Jing, Yongliang Shen, Yangyi Shen, Qiuyu Wang, Hao Ouyang, Hao Chen, and Chunhua Shen. Time is a feature: Exploiting temporal dynamics in diffusion language models. *arXiv preprint arXiv:2508.09138*, 2025.

Qingyan Wei, Yaojie Zhang, Zhiyuan Liu, Dongrui Liu, and Linfeng Zhang. Accelerating diffusion large language models with slowfast sampling: The three golden principles. *arXiv preprint arXiv:2506.10848*, 2025.

Chengyue Wu, Hao Zhang, Shuchen Xue, Zhijian Liu, Shizhe Diao, Ligeng Zhu, Ping Luo, Song Han, and Enze Xie. Fast-dllm: Training-free acceleration of diffusion llm by enabling kv cache and parallel decoding. *arXiv preprint arXiv:2505.22618*, 2025.

Jingxuan Wu, Zhenglin Wan, Xingrui Yu, Yuzhe Yang, Yiqiao Huang, Ivor Tsang, and Yang You. Time-annealed perturbation sampling: Diverse generation for diffusion language models. *arXiv preprint arXiv:2601.22629*, 2026.

Jiacheng Ye, Shansan Gong, Liheng Chen, Lin Zheng, Jiahui Gao, Han Shi, Chuan Wu, Xin Jiang, Zhenguo Li, Wei Bi, et al. Diffusion of thought: Chain-of-thought reasoning in diffusion language models. *Advances in Neural Information Processing Systems*, 37:105345–105374, 2024.

Jiacheng Ye, Zhihui Xie, Lin Zheng, Jiahui Gao, Zirui Wu, Xin Jiang, Zhenguo Li, and Lingpeng Kong. Dream 7b: Diffusion large language models. *arXiv preprint arXiv:2508.15487*, 2025.

Shun Zou, Yong Wang, Zehui Chen, Lin Chen, Chongyang Tao, Feng Zhao, and Xiangxiang Chu. Breaking block boundaries: Anchor-based history-stable decoding for diffusion large language models. *arXiv preprint arXiv:2604.08964*, 2026.

A Related Works

Diffusion Language Models While autoregressive (AR) models dominate natural language generation, their strict left-to-right causal nature restricts bidirectional context modeling and suffers from error propagation. Diffusion Language Models (DLMs) have emerged as a compelling alternative, evolving from early continuous formulations [Han et al., 2023, Dieleman et al., 2022] to discrete Masked Diffusion Models [Saharia et al., 2022, Austin et al., 2021a]. Recently, large-scale DLMs such as LLaDA [Nie et al., 2025] and Dream [Ye et al., 2025] have demonstrated generative and reasoning performance on par with canonical AR baselines.

Fundamentally, DLMs diverge from AR models in their decoding paradigm. Instead of sequential token prediction, DLMs formulate generation as an iterative and bidirectional mask-infilling process [Ye et al., 2024, Li et al., 2025c]. By globally refining a fully masked sequence over discrete timesteps, DLMs naturally support parallel decoding. However, this introduces a severe efficiency-accuracy trade-off: unmasking multiple tokens simultaneously accelerates inference but often degrades semantic quality due to the loss of joint dependency modeling [Hu et al., 2025, Wu et al., 2025]. While existing literature predominantly focuses on architectural scaling or training objectives [Gong et al., 2024, Bie et al., 2025], our work explicitly targets the underexplored bottleneck of high-quality decoding to fully unleash the intrinsic reasoning capabilities of DLMs during the iterative generation phase.

Test-Time Decoding and Guidance Recent studies on DLMs extensively explore test-time decoding strategies to optimize generation quality, diversity, and computational efficiency. To address the critical challenge of token commitment timing, researchers have theoretically validated the efficacy of confidence-based decoding [Cai and Li, 2026]. Practical methods, such as KLASS [Kim et al.] and Prophet [Li et al., 2025b], utilize entropy bounds or KL-divergence to dictate unmasking schedules, while hybrid approaches dynamically alternate between beam search and rapid sampling to balance exploration and computational overhead [Cao et al.]. Capitalizing on the non-autoregressive nature of DLMs, frameworks including Dynamic Sliding Block scheduling Luo et al. [2026] and SlowFast sampling [Wei et al., 2025] modulate decoding step sizes to accelerate parallel generation without compromising output quality. Advancing inference-time scaling and reasoning, techniques like trajectory refinement [Dang et al., 2025] and time-annealed perturbation sampling [Wu et al., 2026] significantly boost generation fidelity. Additionally, to mitigate the myopic assumptions of standard greedy decoding, lookahead mechanisms [Mo et al., 2025] are introduced to foresee future states and guide current decisions. Finally, reliability-enhancing paradigms from autoregressive models, such as Self-Consistency and semantic entropy [Wang et al., 2025], have been successfully adapted to DLMs. Integrated with test-time heuristics like inference-time remasking [Huang et al., 2025], these strategies effectively reduce hallucinations and improve the temporal stability of generated outputs.

Beyond optimizing decoding schedules and unmasking heuristics, inference-time guidance has emerged as a pivotal direction for DLMs, enabling generation steering toward desired attributes without retraining. Originating from continuous diffusion models [Ho and Salimans, 2022] and subsequently adapted to continuous language embeddings [Li et al., 2022], Classifier-Free Guidance (CFG) and its variants have recently been extensively explored in discrete DLM settings. Training-free paradigms have rapidly evolved, encompassing contrastive guided decoding via learned classifiers [Lovelace et al., 2024], posterior sampling for hard lexical constraints [Cardei et al.], and iterative logit refinement [Avrahami and Nachmani, 2026]. To ensure faithful sampling, techniques like Sequential Monte Carlo debiasing [Lee et al., 2025] have been introduced to address the bias of approximate guidance in discrete spaces, while Reward-Free Guidance [Chen et al., 2025] leverages implicit value estimates from the denoising model itself to enhance reasoning without external reward models.

Despite these advancements, standard CFG in discrete DLMs typically relies on interpolating conditional predictions with a static unconditional baseline (e.g., a fully masked sequence), which often depends on superficial confidence scores and fails to adapt to dynamic certainty shifts across token positions during iterative refinement. Recent efforts to address this limitation include adaptive guidance strategies that modulate CFG strength based on per-token uncertainty estimates [Li et al., 2025a], yet these approaches remain constrained by the fundamental paradigm of contrasting conditional against a fixed unconditioned state. Addressing this critical gap, we propose a novel approach that leverages a core reasoning unit to provide dynamic, step-aware guidance throughout the denoising trajectory.

B More Implementation Details

B.1 Datasets

We provide detailed descriptions of the datasets used in our experiments as follows:

- **GSM8K** [Cobbe et al., 2021] is a benchmark consisting of 8,500 high-quality grade school math word problems, requiring multi-step arithmetic reasoning to solve. Each problem is annotated with a detailed natural language solution, making it widely adopted for evaluating the mathematical reasoning capabilities of language models.
- **MATH500** [Hendrycks et al., 2021] is a challenging mathematics benchmark sampled from the MATH dataset, spanning 500 competition-level problems across seven subject areas including algebra, geometry, and number theory. It is designed to assess advanced mathematical problem-solving ability and is considered significantly more difficult than GSM8K.
- **HumanEval** [Chen et al., 2021] is a code generation benchmark comprising 164 hand-crafted Python programming problems, each accompanied by a function signature, docstring, and a set of unit tests. Model-generated solutions are evaluated via functional correctness through test case execution.
- **MBPP** [Austin et al., 2021b] (Mostly Basic Python Problems) consists of approximately 500 crowd-sourced Python programming tasks designed for entry-level programmers. Each problem includes a natural language description, a reference solution, and automated test cases for evaluation.
- **SVAMP** [Patel et al., 2021] is a challenge dataset for arithmetic word problems, constructed by applying controlled structural variations to existing problems. It is specifically designed to test the robustness of models against simple textual perturbations that should not affect the underlying mathematical reasoning.
- **Countdown** [Gandhi et al., 2024] is a combinatorial arithmetic reasoning task in which the model is required to combine a given set of numbers using basic arithmetic operations to reach a specified target value. Its combinatorially large search space makes it a particularly demanding benchmark for evaluating structured reasoning and search capabilities.

B.2 Implementation Details

Our implementation builds upon the official codebases of LLaDA[1] and Dream[2], and all evaluations are conducted using the `lm-evaluation-harness` framework[3] to ensure consistency and reproducibility across benchmarks. While this framework provides variance metrics for the evaluations, we empirically observed that the variance remains approximately consistent across different methods; therefore, we report only the mean performance scores for clarity. All experiments are performed on NVIDIA H800 GPUs (80GB memory), with an average inference latency of approximately 10 seconds per sample.

Evaluation Tasks and Settings. We evaluate our method on six benchmarks spanning three domains: mathematical reasoning (GSM8K and MATH500), code generation (HumanEval and MBPP), and logical reasoning (SVAMP and Countdown). For GSM8K and MATH500, we adopt a zero-shot chain-of-thought prompting strategy. For HumanEval and MBPP, model performance is assessed via functional correctness through automated test case execution.

Hyperparameter Settings. For all main experiments on LLaDA, the sampling temperature is set to 0, and the `low_confidence` remasking strategy is employed. We set the EMA decay factor to 0.9 and use $k = 256$ for the Top-K approximation in JS divergence computation. By default, the block length is 32, the guidance scale (CFG scale) is 0.3, and the Top-K sampling value is fixed at 8 across all tasks. Generation lengths are configured per benchmark, with the number of denoising steps

[1] https://github.com/ML-GSAI/LLaDA

[2] https://github.com/DreamLM/Dream

[3] https://github.com/EleutherAI/lm-evaluation-harness

Algorithm 1 FoCore: Focus on the Core

Require: Model f_θ, prompt X_0, generation length L, block length B, steps per block T, guidance scale ω, Top-K, EMA decay α
Ensure: Generated sequence X

Initialize $X \leftarrow [X_0; \texttt{[MASK]}^L]$
Initialize $S^{(i)} \leftarrow 0$, $P_{\text{prev}}^{(i)} \leftarrow \mathbf{0}$ for all positions i
for each block $b = 1, \ldots, \lfloor L/B \rfloor$ **do**
  for each step $t = 1, \ldots, T$ **do**
    $\mathcal{U}_t \leftarrow \{i : X_i \neq \texttt{[MASK]}\}$ ▷ Unmasked token indices
    // Step 1: Compute conditional logits and distributions
    $L_{\text{cond}} \leftarrow f_\theta(X_t)$
    $P_t(x_i) \leftarrow \text{Softmax}(L_{\text{cond}}[i])$ for all i
    // Step 2: Compute Top-K truncated JS divergence
    for each position $i \in \mathcal{U}_t$ **do**
      $\mathcal{V}_K \leftarrow \text{Top-}K\big(P_t(x_i)\big)$
      $M(v) \leftarrow \frac{1}{2}\big(P_t(v) + P_{t-1}(v)\big), \forall v \in \mathcal{V}_K$
      $D_t^{(i)} \leftarrow \frac{1}{2}\sum_{v\in\mathcal{V}_K} P_t(v)\log\frac{P_t(v)}{M(v)} + \frac{1}{2}\sum_{v\in\mathcal{V}_K} P_{t-1}(v)\log\frac{P_{t-1}(v)}{M(v)}$
    end for
    // Step 3: Update EMA instability scores
    for each position $i \in \mathcal{U}_t$ **do**
      $S_t^{(i)} \leftarrow \alpha \cdot S_{t-1}^{(i)} + (1-\alpha)\cdot D_t^{(i)}$
    end for
    // Step 4: Identify HD tokens and construct negative sample
    $\epsilon \sim \mathcal{U}(0, 10^{-8})$ ▷ Tie-breaking noise
    $\mathcal{K}_t \leftarrow \arg\max_{\mathcal{K}\subset\mathcal{U}_t, |\mathcal{K}|=K} \sum_{j\in\mathcal{K}} \left(S_t^{(j)} + \epsilon\right)$
    $(\tilde{X}_t)_j \leftarrow \texttt{[MASK]}$ if $j \in \mathcal{K}_t$, else $(\tilde{X}_t)_j \leftarrow (X_t)_j$
    // Step 5: Self-Contrast
    $L_{\text{uncond}} \leftarrow f_\theta(\tilde{X}_t)$
    $\hat{L}_t \leftarrow L_{\text{uncond}} + (\omega+1)\cdot\big(L_{\text{cond}} - L_{\text{uncond}}\big)$
    // Step 6: Decode tokens based on guided logits
    Update X_t by unmasking high-confidence tokens via $\hat{L}_t$
    $P_{\text{prev}} \leftarrow P_t$ ▷ Cache distributions for next step
  end for
end for
return X

strictly matching the generation length. For FoCore_A, the early-exit threshold is set to $\tau = 0.01$, and the number of parallel decoding tokens, m, ranges from 1 to 10, tailored to the specific characteristics of each benchmark.

Baselines. We benchmark our proposed method against a diverse set of strong baselines. The first group consists of standard decoding strategies, including Random Sampling, Greedy Decoding, Confidence-based Decoding, and Standard Classifier-Free Guidance (Std CFG). The second group comprises recent state-of-the-art decoding methods, specifically A-CFG [Li et al., 2025a], KLASS [Kim et al.], and Prophet [Li et al., 2025b]. All baselines are reproduced using their original hyperparameter configurations to ensure a fair comparison.

B.3 License of models and data

All models and datasets used in this work are publicly available for academic research. Their references and specific licenses are summarized below:

Models:

- **LLaDA** [Bie et al., 2025] : Apache License 2.0.

Algorithm 2 FoCore_A: FoCore: Focus on the Core Acceleration

Require: Model f_θ, prompt X_0, generation length L, block length B, steps per block T, guidance scale ω, Top-K, EMA decay α, early-exit threshold τ, parallel decoding count m
Ensure: Generated sequence X

Initialize $X \leftarrow [X_0; \texttt{[MASK]}^L]$
Initialize $S^{(i)} \leftarrow 0$, $P_{\text{prev}}^{(i)} \leftarrow \mathbf{0}$ for all positions i
for each block $b = 1, \ldots, \lfloor L/B \rfloor$ **do**
  `early_exit` $\leftarrow$ `False`
  for each step $t = 1, \ldots, T$ **do**
    if `early_exit` **then**
      break
    end if
    $\mathcal{U}_t \leftarrow \{i : X_i \neq \texttt{[MASK]}\}$
    // Steps 1–5: Same as FoCore (Algorithm 1)
    Compute L_{cond}, P_t, $D_t^{(i)}$, $S_t^{(i)}$, $\mathcal{K}_t$, $\tilde{X}_t$, $\hat{L}_t$
    // Step 6: Monitor mean instability for early-exit
    $\bar{s}_t \leftarrow \frac{1}{|\mathcal{U}_t|} \sum_{j \in \mathcal{U}_t} S_t^{(j)}$
    if $\bar{s}_t < \tau$ **then** ▷ HD tokens have sufficiently converged
      Decode m additional high-confidence tokens in parallel via $\hat{L}_t$
      `early_exit` $\leftarrow$ `True`
    else
      // Normal decoding step
      Update X_t by unmasking high-confidence tokens via $\hat{L}_t$
    end if
    $P_{\text{prev}} \leftarrow P_t$
  end for
end for
return X

- **Dream** [Ye et al., 2025] : Apache License 2.0.

Datasets:

- **GSM8K** [Cobbe et al., 2021]: MIT License.
- **MATH500** [Hendrycks et al., 2021]: MIT License.
- **HumanEval** [Chen et al., 2021]: MIT License.
- **MBPP** [Austin et al., 2021b]: CC-BY 4.0 License.
- **SVAMP** [Patel et al., 2021]: MIT License.
- **Countdown** [Gandhi et al., 2024]: MIT License.

C Pseudo Code

We provide the pseudo code for FoCore (Algorithm 1) and FoCore_A (Algorithm 2).

D Theoretical Analysis

D.1 Preliminary: Notation and Setup

Let $\mathcal{V}$ denote the vocabulary of size $|\mathcal{V}|$, and $\Delta^{|\mathcal{V}|-1}$ the corresponding probability simplex. For a sequence of length N, the predicted distribution at position i and decoding step t is denoted by $P_t^{(i)} \in \Delta^{|\mathcal{V}|-1}$, for $t = 0, 1, \ldots, T$.

Definition D.1 (Positional Conditional Entropy). *Given the ground-truth tokens at all other positions x^*_{-i}, the* intrinsic semantic difficulty *of position i is defined as the conditional entropy:*

$$\mathcal{H}_i \triangleq H(x_i \mid x^*_{-i}) = -\sum_{v \in \mathcal{V}} p(x_i = v \mid x^*_{-i}) \log p(x_i = v \mid x^*_{-i}) \tag{9}$$

Definition D.2 (Formal Characterization of HD/LD Tokens). *Given a threshold $\eta > 0$, we formally define the partition of tokens into hard-to-decode (HD) and easy-to-decode (LD) categories as:*

$$i \in \mathrm{HD} \iff \mathcal{H}_i > \eta; \qquad i \in \mathrm{LD} \iff \mathcal{H}_i \leq \eta \tag{10}$$

D.2 Distribution Instability as a Proxy for Semantic Difficulty

D.2.1 Step-wise JS Divergence Upper-bounded by Conditional Entropy

Proposition D.1 (Entropic Upper Bound on JS Divergence). *Let f_θ be a sufficiently trained masked diffusion language model satisfying $P_t^{(i)} \approx p(x_i \mid x^t)$. Let Δ^t denote the contextual increment introduced at step t, i.e., the set of newly decoded tokens. Then the step-wise JS divergence at position i satisfies:*

$$D_t^{(i)} \leq \frac{1}{2} I(x_i; \Delta^t \mid x^{t-1}) \leq \frac{1}{2}\mathcal{H}_i \tag{11}$$

Proof. **First inequality.** By the standard relationship between JS divergence and KL divergence:

$$D_t^{(i)} = \mathrm{JS}(P_t^{(i)} \| P_{t-1}^{(i)}) \leq \frac{1}{2} D_{\mathrm{KL}}(P_t^{(i)} \| P_{t-1}^{(i)}) \tag{12}$$

Since the distributional shift between $P_t^{(i)}$ and $P_{t-1}^{(i)}$ is induced solely by the contextual increment Δ^t, applying the Data Processing Inequality yields:

$$D_{\mathrm{KL}}(P_t^{(i)} \| P_{t-1}^{(i)}) \leq I(x_i; \Delta^t \mid x^{t-1}) \tag{13}$$

Second inequality. By the definition of conditional mutual information and the monotonicity of conditional entropy (conditioning on more information does not increase entropy):

$$I(x_i; \Delta^t \mid x^{t-1}) \leq H(x_i \mid x^{t-1}) \leq H(x_i \mid x^*_{-i}) = \mathcal{H}_i \tag{14}$$

Combining the two inequalities completes the proof. □

Corollary D.1 (Inherent Stability of LD Tokens). *If $i \in \mathrm{LD}$, then $\mathcal{H}_i \leq \eta$, and consequently:*

$$D_t^{(i)} \leq \frac{\eta}{2} \approx 0 \tag{15}$$

That is, the predictive distribution of an LD token remains stable under arbitrary contextual changes, with near-zero step-wise JS divergence. Conversely, for HD tokens ($\mathcal{H}_i > \eta$), the upper bound is looser, permitting larger step-wise divergence.

Proposition D.1 and Corollary D.1 jointly provide a principled theoretical justification for the use of distributional instability as a proxy for identifying HD tokens: the observed distributional instability is a necessary consequence of high conditional entropy.

D.2.2 Approximation Error of Top-K Truncation

FoCore employs a Top-K truncated JS divergence to reduce computational overhead, restricting the computation to the K highest-probability tokens under the current step's distribution. Let $\widetilde{D}_t^{(i)}$ denote the JS divergence computed over the full vocabulary, and $D_t^{(i)}$ its Top-K truncated counterpart. Define the residual tail mass of the mixture distribution as:

$$\epsilon_K = 1 - \sum_{v \in \mathcal{V}_K} M(v), \quad M(v) = \frac{1}{2}\big(P_t(v) + P_{t-1}(v)\big) \tag{16}$$

Proposition D.2 (Truncation Approximation Error Bound).

$$\left| \widetilde{D}_t^{(i)} - D_t^{(i)} \right| \leq \log 2 \cdot \epsilon_K \tag{17}$$

Proof. By the boundedness of JS divergence, $0 \le \mathrm{JS}(\cdot\|\cdot) \le \log 2$. The contribution of the truncated token set $\mathcal{V} \setminus \mathcal{V}_K$ to the full JS divergence is upper bounded by:

$$\log 2 \cdot \sum_{v \notin \mathcal{V}_K} M(v) = \log 2 \cdot \epsilon_K \tag{18}$$

which directly yields the stated bound. □

Remark D.1. In practice, the softmax output distributions of language models are highly concentrated over a small set of high-frequency tokens, resulting in $\epsilon_K \ll 1$. Consequently, the Top-K truncation achieves an $O(|\mathcal{V}|/K)$-fold reduction in computational cost while introducing only an $O(\epsilon_K)$ approximation error, which is negligible in practice.

D.3 HD Tokens Dominate the Decoding Trajectory

D.3.1 HD Tokens as Information Bottlenecks

Definition D.3 (Positional Information Contribution)**.** *The information contribution of position i to the overall sequence is defined as its mutual information with all other positions:*

$$\mathcal{I}_i \triangleq I(x_i;\, x^*_{-i}) = H(x_i) - H(x_i \mid x^*_{-i}) = H(x_i) - \mathcal{H}_i \tag{19}$$

Proposition D.3 (HD Tokens as Information Bottlenecks)**.** *For any subset of positions $\mathcal{S} \subset [N] \setminus \{i\}$, the conditional information contribution of position i to the tokens in $\mathcal{S}$ satisfies:*

$$I(x_i;\, x_{\mathcal{S}}) \ge \mathcal{H}_{\mathcal{S}} - \mathcal{H}_{\mathcal{S} \cup \{i\}} \tag{20}$$

where $\mathcal{H}_{\mathcal{S}} = H(x_{\mathcal{S}} \mid x_{[N]\setminus\mathcal{S}})$ denotes the joint conditional entropy of the positions in $\mathcal{S}$.

Proof. By the chain rule of mutual information:

$$I(x_i;\, x_{\mathcal{S}}) = H(x_{\mathcal{S}}) - H(x_{\mathcal{S}} \mid x_i) \tag{21}$$

For an HD token at position i, the marginal entropy $H(x_i)$ is large (high prior uncertainty) and $\mathcal{H}_i > \eta$ (high intrinsic difficulty), implying that the realization of x_i exerts a significant influence on the predictive uncertainty of other positions:

$$H(x_{\mathcal{S}} \mid x_i = v_1) \ne H(x_{\mathcal{S}} \mid x_i = v_2), \quad \forall\, v_1 \ne v_2 \tag{22}$$

By the convexity of conditional entropy and the non-negativity of mutual information, the reduction in uncertainty over $\mathcal{S}$ upon observing x_i, namely $\mathcal{H}_{\mathcal{S}} - \mathcal{H}_{\mathcal{S}\cup\{i\}}$, constitutes a lower bound on $I(x_i;\, x_{\mathcal{S}})$. □

Proposition D.3 establishes that HD tokens are critical semantic constraint nodes in the sequence: their predictive uncertainty propagates through mutual information to all other positions, rendering the correct decoding of HD tokens an *information bottleneck* that governs the overall generation quality.

D.3.2 Variance Propagation: HD Tokens Induce Path-level Perturbation

Theorem D.1 (Path-level Dominance of HD Tokens)**.** *Assume the denoising process of the masked diffusion model satisfies the Markov property, i.e., $P_t^{(j)} = [f_\theta(x^t)]_j$, and that f_θ is Lipschitz continuous with constant $C > 0$ with respect to its input. For position $j \ne i$, define the* path sensitivity *of its predictive distribution to the token at position i as:*

$$\Gamma_t^{(i\to j)} \triangleq \mathbb{E}_{v_1, v_2 \sim P_t^{(i)}}\left[\left\| P_t^{(j)}\big|_{x_i = v_1} - P_t^{(j)}\big|_{x_i = v_2} \right\|_1\right] \tag{23}$$

Then there exists a constant $C' > 0$ such that:

$$\Gamma_t^{(i\to j)} \le C' \cdot \sqrt{\mathcal{H}_i} \tag{24}$$

and the cumulative path perturbation across all other positions satisfies:

$$\sum_{j \ne i} \Gamma_t^{(i\to j)} \le C'(N-1) \cdot \sqrt{\mathcal{H}_i} \tag{25}$$

Proof. The variation in the predictive distribution at position j is induced by the change in context arising from different realizations of x_i. By the Lipschitz continuity of f_θ:

$$\left\| P_t^{(j)}\big|_{x_i=v_1} - P_t^{(j)}\big|_{x_i=v_2} \right\|_1 \leq C \cdot \|E(v_1) - E(v_2)\|_2 \tag{26}$$

Taking expectations and applying Jensen's inequality:

$$\Gamma_t^{(i\to j)} \leq C \cdot \mathbb{E}_{v_1,v_2}[\|E(v_1) - E(v_2)\|_2] \leq C \cdot \sqrt{2\,\mathrm{Var}_{P_t^{(i)}}[E(v)]} \tag{27}$$

Since the variance of the embedding under distribution $P_t^{(i)}$ is bounded by the distributional spread, which is in turn upper bounded by the entropy as $\sqrt{2H(P_t^{(i)})} \leq \sqrt{2\mathcal{H}_i}$, setting $C' = C\sqrt{2}$ establishes the first inequality. Summing over all $j \neq i$ immediately yields the cumulative bound. □

Corollary D.2 (Conditional Entropy Governs Path Dominance)**.**

$$\mathcal{H}_i \nearrow \implies \sum_{j\neq i} \Gamma_t^{(i\to j)} \nearrow \tag{28}$$

That is, the higher the conditional entropy of position i, the larger its cumulative perturbation on the predictive distributions of all other positions along the decoding trajectory. This theoretically establishes HD tokens as the dominant control nodes of the decoding path.

D.3.3 Self-Contrast as Entropy Reduction on HD Tokens

Theorem D.2 (Self-Contrast Reduces Output Entropy of HD Tokens)**.** *The self-contrastive logits in FoCore are constructed as:*

$$\hat{L}_t = L_{\text{cond}} + \omega \cdot (L_{\text{cond}} - L_{\text{uncond}}) \tag{29}$$

which is equivalent in log-probability space to:

$$\log \hat{p}(x_i \mid X_t) \propto (1+\omega) \log p_\theta(x_i \mid X_t) - \omega \log p_\theta(x_i \mid \tilde{X}_t) \tag{30}$$

For HD token positions $i \in \mathcal{K}_t$, the self-contrast operation strictly reduces the entropy of the output distribution:

$$H(\hat{p}(\cdot \mid X_t)) \leq H(p_\theta(\cdot \mid X_t)) - \omega \cdot I_t^{(i)} + O(\omega^2) \tag{31}$$

where $I_t^{(i)} = D_{\mathrm{KL}}\Big(p_\theta(x_i \mid X_t) \,\|\, p_\theta(x_i \mid \tilde{X}_t)\Big) \geq 0$ denotes the information gap induced by masking the HD tokens.

Proof. Let $\ell(v) = \log p_\theta(v \mid X_t)$ and $\tilde{\ell}(v) = \log p_\theta(v \mid \tilde{X}_t)$ denote the conditional and unconditional logits, respectively. The self-contrastive logit is:

$$\hat{\ell}(v) = (1+\omega)\ell(v) - \omega\tilde{\ell}(v) \tag{32}$$

with the corresponding self-contrastive distribution $\hat{p}(v) \propto \exp(\hat{\ell}(v))$.

Expanding the output entropy $H(\hat{p})$ via a first-order Taylor expansion around $\omega = 0$:

$$H(\hat{p}) = H(p_\theta(\cdot \mid X_t)) - \omega \cdot \frac{d}{d\omega} H(\hat{p})\bigg|_{\omega=0} + O(\omega^2) \tag{33}$$

Computing the first-order derivative:

$$\frac{d}{d\omega} H(\hat{p})\bigg|_{\omega=0} = \sum_v p_\theta(v \mid X_t)\left[\ell(v) - \tilde{\ell}(v)\right] = \mathbb{E}_{p_\theta}\left[\log \frac{p_\theta(v \mid X_t)}{p_\theta(v \mid \tilde{X}_t)}\right] = I_t^{(i)} \geq 0 \tag{34}$$

Substituting back yields the stated bound. □

Corollary D.3 (Self-Contrast Compresses Path-level Perturbation)**.** *Combining Theorem D.1 and Theorem D.2, the self-contrast mechanism reduces the effective conditional entropy of HD token i from $\mathcal{H}_i$ to:*

$$\hat{\mathcal{H}}_i \approx \mathcal{H}_i - \omega \cdot I_t^{(i)} \tag{35}$$

Consequently, the cumulative path perturbation upper bound is reduced from $C'(N-1)\sqrt{\mathcal{H}_i}$ to $C'(N-1)\sqrt{\hat{\mathcal{H}}_i}$, thereby theoretically quantifying the improvement in decoding trajectory stability achieved by FoCore.

D.4 Summary

The theoretical analysis above establishes a complete and closed mathematical foundation for FoCore, forming the following causal chain:

$$\underbrace{\mathcal{H}_i > \eta}_{\text{HD token definition}} \xRightarrow{\text{Prop. D.1}} \underbrace{D_t^{(i)} \lesssim \tfrac{1}{2}\mathcal{H}_i}_{\text{instability detectable}} \xRightarrow{\text{Thm. D.1}} \underbrace{\textstyle\sum_j \Gamma_t^{(i \to j)} \leq C'(N-1)\sqrt{\mathcal{H}_i}}_{\text{dominates decoding path}} \xRightarrow{\text{Thm. D.2}} \underbrace{\hat{\mathcal{H}}_i < \mathcal{H}_i}_{\text{uncertainty compressed}} \tag{36}$$

Specifically: **(1)** Proposition D.1 establishes from an information-theoretic perspective that distributional instability is a reliable estimator of conditional entropy, validating the use of step-wise JS divergence for HD token identification. **(2)** Proposition D.2 guarantees that the approximation error introduced by Top-K truncation is theoretically controllable and negligible in practice. **(3)** Proposition D.3 and Theorem D.1 jointly establish that HD tokens dominate the entire decoding trajectory through mutual information propagation, with their influence scaling as $O(\sqrt{\mathcal{H}_i})$. **(4)** Theorem D.2 quantifies the extent to which the FoCore self-contrast mechanism compresses the predictive uncertainty of HD tokens, thereby providing a theoretical guarantee for the improvement in decoding trajectory stability.

E More Experiments

E.1 SVAMP and Countdown datasets

Table 4 presents the performance comparison of FoCore against baseline methods on the SVAMP and Countdown benchmarks under two token budget settings (128 and 256). Overall, **FoCore demonstrates superior and consistent performance across both tasks and budget configurations.** Specifically, on the Countdown benchmark, FoCore achieves the highest accuracy among all methods, attaining 21.8 and 18.8 under the 128 and 256 token budgets, respectively, outperforming CFG (20.8/16.8), A-CFG (17.6/19.2), and Hybrid (20.4/15.2) by a notable margin. On the SVAMP benchmark, although A-CFG and Hybrid marginally surpass FoCore under their respective optimal budget settings, their performance exhibits considerable instability across different token budgets. For instance, Hybrid achieves only 86.6 under the 128 token budget while rising to 89.0 under 256 tokens, indicating a strong sensitivity to budget variation. In contrast, FoCore maintains competitive and stable results of 88.0 and 87.0 on SVAMP under the 128 and 256 token budgets, respectively, demonstrating greater robustness to token budget changes. These results suggest that FoCore strikes a better balance between *task performance* and *budget robustness*, making it a more reliable approach across diverse reasoning scenarios.

Table 4: Performance comparison of different methods on SVAMP and Countdown benchmarks under different token budget settings (128 and 256).

Method	**SVAMP**		**Countdown**	
	128	**256**	**128**	**256**
CFG	86.3	86.6	20.8	16.8
A-CFG	**90.0**	86.6	17.6	18.2
Hybrid	86.6	**89.0**	20.4	15.2
FoCore (Ours)	<u>88.0</u>	<u>87.0</u>	**21.8**	**18.8**

E.2 Effect of Parallel Token Budget m on the Speedup–Accuracy Trade-off

To investigate the impact of the parallel token budget m (i.e., the number of additional high-confidence tokens decoded in parallel at each step) on both decoding efficiency and task performance, we conduct an ablation study on a randomly sampled subset of 200 examples from the GSM8K test set. We vary m from 0 to 14 and record the average generation time per sample, average speedup, and task accuracy for each configuration. The case of $m = 0$ serves as the baseline, corresponding to standard sequential decoding without any parallel token prediction. The results are summarized in Table 5,

Table 5: Effect of parallel token budget m on decoding efficiency and task accuracy. Accuracy is evaluated on a random subset of 200 samples from the GSM8K test set.

m	Avg Time/Sample (s)	Avg Speedup	Accuracy (%)
0	8.14	1.00×	73.00
1	7.68	1.03×	73.50
2	6.66	1.06×	74.50
3	6.41	1.10×	73.50
4	6.20	1.14×	71.00
5	6.00	1.18×	**75.50**
6	5.77	1.23×	72.50
7	5.55	1.28×	71.50
8	5.32	1.33×	70.50
9	5.09	1.39×	69.50
10	4.87	1.45×	63.00
11	4.64	1.52×	57.50
12	4.46	1.60×	57.00
13	4.24	1.68×	49.00
14	4.02	1.78×	49.00

from which we draw the following observations.

Speedup increases monotonically with m. As m increases, the number of tokens decoded in parallel at each step grows, leading to a steady reduction in average generation time from 8.14s at m=0 to 4.02s at m=14, with the corresponding speedup rising from $1.00\times$ to $1.78\times$. This confirms that the parallel decoding mechanism effectively reduces the total number of forward passes required during generation.

Accuracy remains stable for small m and degrades sharply beyond a threshold. Within the range $m \leq 9$, task accuracy remains largely stable, fluctuating between 69.50% and 75.50% and staying broadly comparable to the baseline of 73.00%. This plateau suggests that once a high-confidence (HD) token is identified as a reliable anchor, the locally neighboring low-confidence (LD) tokens in its vicinity remain semantically stable, making it safe to decode them in parallel without degrading generation quality. However, once $m \geq 10$, accuracy drops sharply, falling from 69.50% at m=9 to 63.00% at m=10, and further declining to 49.00% at m=14. This indicates that excessively extending the parallel decoding window pushes prediction beyond the locally stable region: the tokens being forced into parallel decoding no longer carry sufficient confidence to support reliable predictions, thereby introducing erroneous tokens into the generation sequence and substantially corrupting the reasoning chain.

Recommended operating range. Balancing efficiency and accuracy, $m \in [5, 9]$ constitutes a favorable operating region, yielding speedups of $1.18\times$–$1.39\times$ while preserving accuracy at or above the sequential baseline. In our main experiments, we adopt m=10, which provides a $1.45\times$ speedup and represents an efficiency-oriented configuration chosen with the benefit of full-dataset statistical stability, where FoCore_A retains competitive performance (GSM8K: 73.99%).

Overall, these results highlight m as a critical hyperparameter governing the efficiency–accuracy trade-off in parallel decoding. The semantic stability of local LD tokens conditioned on HD token anchors provides a principled justification for moderate parallel decoding, while excessive extension of the parallel window beyond this stable region leads to significant quality degradation. We recommend selecting m within the range $[5, 10]$ in practice, depending on the accuracy sensitivity of the target task.

F Case Study

F.1 Case:1

As shown in Table 6, we present a qualitative comparison between the baseline method (**Confidence**) and our proposed method (**FoCore**) on a representative GSM8K arithmetic reasoning sample. Both methods take the same input prompt, yet produce markedly different outputs.

The **Confidence** baseline generates a fluent but incorrect response, arriving at \$48 as the final answer. From the volatility analysis, several tokens exhibit high information density, including `second` (JS $= 0.094$), `costs` (JS $= 0.207$), `price` (JS $= 0.128$), and notably the discount factor token `0` (JS $= 5.000$). However, since the Confidence baseline selects tokens solely based on prediction confidence, it fails to recognise and attend to the critical conditional token `every second`. As a result, the model overlooks the key constraint that only every second glass is eligible for the discount, and incorrectly applies the discounted price to all 16 glasses ($16 \times \$3 = \$48$), causing the reasoning path to deviate from the correct direction from the very beginning.

FoCore, by contrast, explicitly identifies high information density (HD) tokens during the decoding process and treats them as key anchors for subsequent reasoning. Specifically, FoCore detects high JS-divergence tokens such as `second`, `costs`, `price`, and `0`, and recognizes that the core of the discount condition lies in `every second`, i.e., the discount applies to only every other glass. Guided by these HD tokens, the model's reasoning path is correctly directed: the 16 glasses are split into 8 full-price glasses ($8 \times \$5 = \$40$) and 8 discounted glasses ($8 \times \$3 = \$24$), yielding the correct final answer of \$64. This example demonstrates that by perceiving and leveraging HD tokens during decoding, FoCore effectively captures the key constraints of the problem and steers the model toward a correct reasoning path from the outset.

F.2 Case:2

As shown in Table 7, we present a qualitative comparison between the baseline method (**Confidence**) and our proposed method (**FoCore**) on a second GSM8K arithmetic reasoning sample. Both methods take the same input prompt, yet produce markedly different outputs.

The **Confidence** baseline generates an incorrect response, arriving at 13 liters as the final answer. From the volatility analysis, several tokens exhibit high information density, including `water` (JS $= 0.273$), `drink` (JS $= 0.262$), and the intermediate result token `4` (JS $= 0.172$). However, since the Confidence baseline selects tokens solely based on prediction confidence, it fails to recognise and attend to the critical constraint that one liter of orange drink is spilled during pouring. As a result, the model directly computes the water content using the full 10 liters of orange drink ($10 \times \frac{2}{3} = 4$ liters), completely overlooking the quantity change caused by the spill, and incorrectly concludes that there are $4 + 9 = 13$ liters of water in the remaining mixture.

FoCore, by contrast, explicitly identifies high information density (HD) tokens during the decoding process and treats them as key anchors for subsequent reasoning. Specifically, FoCore detects high JS-divergence tokens such as `water` and `drink`, recognising that the computation of liquid quantities is the core information of this problem. Guided by these HD tokens, the model's attention is directed toward the critical constraint "spill 1 liter", correctly adjusting the remaining orange drink to $10 - 1 = 9$ liters before computing its water content ($9 \times \frac{2}{3} = 6$ liters). Combined with the water content of the pineapple drink ($15 \times \frac{3}{5} = 9$ liters), FoCore yields the correct final answer of $6 + 9 = 15$ liters. This example further demonstrates that by perceiving and leveraging HD tokens during decoding, FoCore effectively captures critical constraints embedded in the problem and steers the model toward a correct reasoning path from the outset.

G Limitations

Despite its effectiveness, our work has several limitations. First, although FoCore is training-free, its optimal hyperparameters exhibit slight task-specific variations, increasing the adaptation cost for novel scenarios. Second, while FoCore_A substantially accelerates generation by reducing overall decoding steps, it inherits the per-step computational burden of the Classifier-Free Guidance (CFG) paradigm; thus, our inference latency reductions are strictly established relative to CFG-based

Prompt	Kylar went to the store to buy glasses for his new apartment. One glass **costs** $5, but every **second** glass **costs** only 60% of the **price**. Kylar wants to buy 16 glasses. How much does he need to pay for them?
Confidence	**Every second** glass **costs** 60% of the **price**, so it **costs** $\$5 \times 0.6 = \$3$. Kylar wants to buy 16 glasses, so he needs to pay $16 \times \$3 = \$48$. The final **result** is 48. **Answer: $48**
FoCore	**Every second** glass **costs** 60% of the **price**, so it **costs** $\$5 \times 0.6 = \$3$. Kylar wants to buy 16 glasses, so he will buy $16 \div 2 = 8$ glasses at full **price** and 8 glasses at the discounted **price**. For the full-**price** glasses, he needs to pay $8 \times \$5 = \$40$. For the discounted glasses, he needs to pay $8 \times \$3 = \$24$. The final **result** is $\$40 + \$24 = \$64$. **Answer: $64**

Table 6: Qualitative analysis on a GSM8K sample. HD tokens are colour-coded by JS divergence: **extremely high** (>1.0) , **high** (0.1 – 1.0) , **medium** (0.01 – 0.1) , low (<0.01) .

Prompt	I have 10 **liters** of orange **drink** that are two-thirds **water** and I wish to add it to 15 **liters** of pineapple **drink** that is three-fifths **water**. But as I pour it, I spill 1 **liter** of the orange **drink**. How much **water** is in the remaining 24 **liters**?
Confidence	The orange **drink** has 10 **cdot** $2/3 = 4$ **liters** of **water**. The pineapple **drink** has 15 **cdot** $3/5 = 9$ **liters** of **water**. Thus, there are $4 + 9 = 13$ **liters** of **water**. **Answer: 13**
FoCore	After spilling 1 **liter**, the orange **drink** remaining is $10 - 1 = 9$ **liters**. The orange **drink** has 9 **cdot** $2/3 = 6$ **liters** of **water**. The pineapple **drink** has 15 **cdot** $3/5 = 9$ **liters** of **water**. Thus, there are $6 + 9 = 15$ **liters** of **water**. **Answer: 15**

Table 7: Qualitative analysis on a GSM8K sample. HD tokens are colour-coded by JS divergence: **extremely high** (>1.0) , **high** (0.1 – 1.0) , **medium** (0.01 – 0.1) , low (<0.01) .

baselines. Finally, our current evaluations are focused on 7B-8B parameter backbones (LLaDA and Dream), leaving the validation of our findings on larger-scale models for future work.